Anisotropy Analysis in 3D Gaussian Splatting Project Report

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1 Introduction

Novel-view synthesis, the task of rendering a 3D scene from previously unseen camera positions given a set of reference photographs, has undergone rapid transformation in recent years. Neural Radiance Fields (NeRF) [6] established a compelling baseline by optimising a Multi-Layer Perceptron to represent continuous volumetric density and colour, enabling photorealistic rendering at the cost of slow training and inference. Subsequent work sought to close this speed gap: Plenoxels [2] replaced the MLP with a sparse voxel grid, while Instant Neural Graphics Primitives (InstantNGP) [7] introduced multiresolution hash encoding to achieve interactive training times.

The contribution of Kerbl et al. [3], 3D Gaussian Splatting (3DGS), pushed the frontier further by representing the scene as a collection of explicit, anisotropic 3D Gaussians combined with a tile-based differentiable rasteriser. This representation supports real-time rendering (≥ 100 fps at 1080p) and training times competitive with the fastest NeRF variants, while matching or surpassing Mip-NeRF360 [1] in perceptual quality on established benchmarks.

A defining feature of 3DGS is anisotropic covariance: each Gaussian has an independent scaling vector $\mathbf{s} \in \mathbb{R}^3$ and rotation quaternion $\mathbf{q}$, parameterising the full covariance matrix $\mathbf{\Sigma} = \mathbf{RSS}^\top\mathbf{R}^\top$. This allows a single Gaussian to stretch along surface tangents and compress along surface normals, capturing fine geometric detail compactly. Whether anisotropic Gaussians are uniformly distributed in scene space or concentrated at geometrically meaningful locations is an empirical question this project addresses.

We investigate these questions on the Train scene from the Tanks & Temples benchmark [4], a retired locomotive surrounded by varied terrain. Our objectives are:

- **O1.** Identify scene properties that justify the expectation of anisotropic structure in the learned Gaussian cloud.
- **O2.** Implement 3DGS and train a converged model on the Train scene.
- **O3.** Characterise the distribution and spatial organisation of anisotropy, and evaluate whether rendering quality is systematically lower in highly anisotropic regions.
- **O4.** (*Bonus*) Find potential algorithmic modifications to improve the representation of anisotropic structures.

2 Preliminaries

2.1 3D Gaussian Splatting

3DGS [3] initialises a set of 3D Gaussians from the sparse Structure-from-Motion (SfM) point cloud produced by COLMAP [8]. Each Gaussian i is characterised by a centre $\boldsymbol{\mu}_i \in \mathbb{R}^3$, a covariance matrix $\mathbf{\Sigma}_i = \mathbf{R}_i\mathbf{S}_i\mathbf{S}_i^\top\mathbf{R}_i^\top$ (where $\mathbf{S}_i = \mathrm{diag}(s_{i,0}, s_{i,1}, s_{i,2})$ and $\mathbf{R}_i$ is encoded by quaternion $\mathbf{q}_i$), an opacity $\alpha_i \in [0, 1)$, and spherical harmonic (SH) coefficients encoding view-dependent colour. The Gaussian evaluates as

$$G(\mathbf{x}) = \exp\Big(-\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top\mathbf{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\Big), \tag{1}$$

and the rendered pixel colour follows the standard α-blending formulation

$$C = \sum_{i \in \mathcal{N}} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \tag{2}$$

where the sum is over depth-sorted Gaussians overlapping the pixel and the product term $\prod_{j=1}^{i-1}(1-\alpha_j)$ is the *transmittance*—in probabilistic terms, the likelihood that light is not occluded by any of the $i-1$ distributions in front of Gaussian i. For rasterisation, each Gaussian is projected to 2D using the EWA splatting framework [10]: $\mathbf{\Sigma}' = \mathbf{J}\mathbf{W}\mathbf{\Sigma}\mathbf{W}^\top\mathbf{J}^\top$, where $\mathbf{W}$ is the viewing transformation and $\mathbf{J}$ is the Jacobian of the affine approximation of the projective transformation.

Optimisation minimises a combined photometric loss:

$$\mathcal{L} = (1-\lambda)\mathcal{L}_1 + \lambda \mathcal{L}_{\text{D-SSIM}}, \quad \lambda = 0.2. \tag{3}$$

An adaptive density control mechanism densifies under-reconstructed regions by cloning Gaussians with large view-space positional gradients, and splits over-reconstructed large Gaussians, allowing the count to grow from a few thousand SfM points to several million primitives.

2.2 Anisotropic Gaussian Primitives

Unlike early radiance-field representations that use isotropic blobs or rely on implicit networks, 3DGS employs anisotropic primitives whose shape is fully determined by the covariance matrix $\mathbf{\Sigma}$. Figure 1 contrasts the three qualitative regimes.

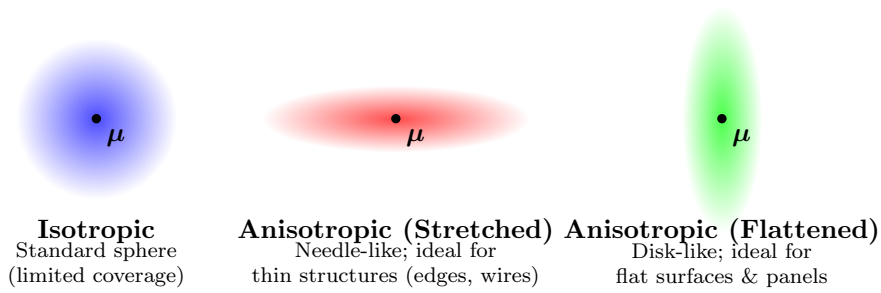

Figure 1: Three qualitative Gaussian shape regimes arising from different covariance matrices $\mathbf{\Sigma}$. Making $\mathbf{\Sigma}$ anisotropic significantly reduces the total number of primitives needed to accurately represent complex geometry.

Each Gaussian carries four groups of optimisable parameters.

- **Position** $\boldsymbol{\mu} \in \mathbb{R}^3$: the 3D centre of the primitive.
- **Covariance** $\mathbf{\Sigma} = \mathbf{R}\mathbf{S}\mathbf{S}^\top\mathbf{R}^\top$: encodes 3D scale $\mathbf{s}$ and rotation $\mathbf{q}$. Anisotropy allows a single primitive to stretch into cigar shapes or flatten like pancakes to cover surfaces efficiently.
- **Opacity** $\alpha \in [0, 1)$: a learnable transparency multiplier applied independently of the spatial Gaussian profile during alpha blending.
- **Spherical Harmonics (SH) coefficients**: instead of a static RGB colour, the colour c_i in Equation (2) is evaluated as a truncated SH expansion $c_i(\mathbf{d}) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} k_{\ell m} Y_\ell^m(\mathbf{d})$, where $\mathbf{d}$ is the unit viewing direction and Y_ℓ^m are real spherical harmonic basis functions. Kerbl et al. use degree $L = 3$ (16 coefficients per colour channel, 48 total), enabling the model to represent complex view-dependent effects such as specular reflections and angle-dependent lighting without an explicit neural network at inference time.

2.3 Adaptive Density Control

The initial Gaussian count is inherited from the SfM point cloud (typically a few thousand points). During training, an adaptive density control (ADC) mechanism dynamically updates the primitive population every 100 iterations. Three complementary operations are employed, illustrated in Figure 2.

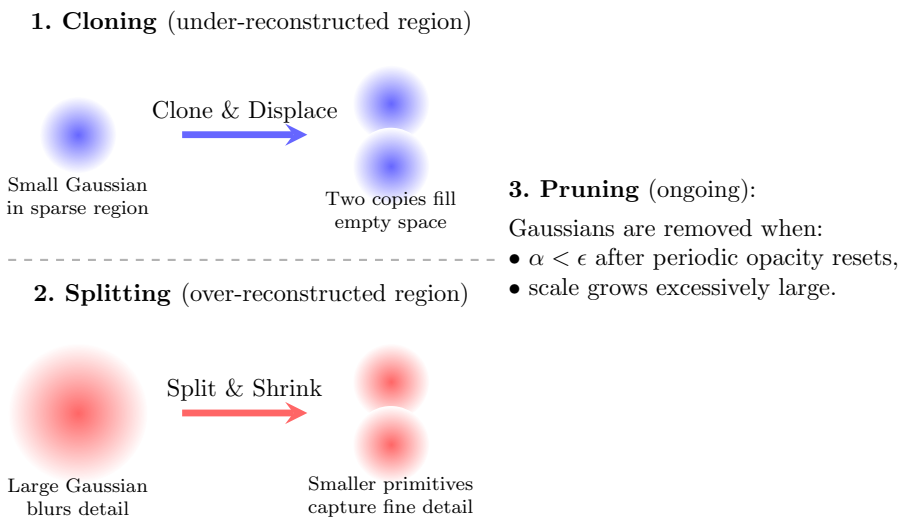

Figure 2: Adaptive density control: (1) cloning duplicates a small Gaussian and displaces the copy into an under-reconstructed region; (2) splitting divides an overly large Gaussian into two smaller ones to capture finer geometry; (3) pruning removes degenerate primitives throughout training.

2.4 Fast Differentiable Tile Rasteriser

Projecting millions of 3D Gaussians onto a 2D sensor ("splatting") must be fast enough to permit real-time feedback during gradient descent. The 3DGS rasteriser achieves this through three coordinated steps, illustrated in Figure 3.

1. **2D projection.** The 3D covariance matrix $\mathbf{\Sigma}$ of each Gaussian is projected to a 2D covariance $\mathbf{\Sigma}'$ via the EWA splatting approximation, yielding a 2D Gaussian "splat" whose footprint depends on the current camera pose.
2. **Tile-based depth sorting.** The screen is partitioned into non-overlapping 16×16 pixel tiles. Each splat is assigned to every tile it overlaps, and the resulting list is sorted front-to-back using a GPU Radix sort—an $O(n)$ algorithm well-suited to massively parallel hardware. This step is performed per-frame during both training and inference.
3. **Alpha blending.** The rasteriser iterates through the sorted splat list and accumulates pixel colours via Equation (2), stopping early once cumulative opacity exceeds a saturation threshold. Crucially, the backward pass stores the sorted order and per-tile Gaussian lists so that gradients flow through the blending operation to update all four parameter groups.

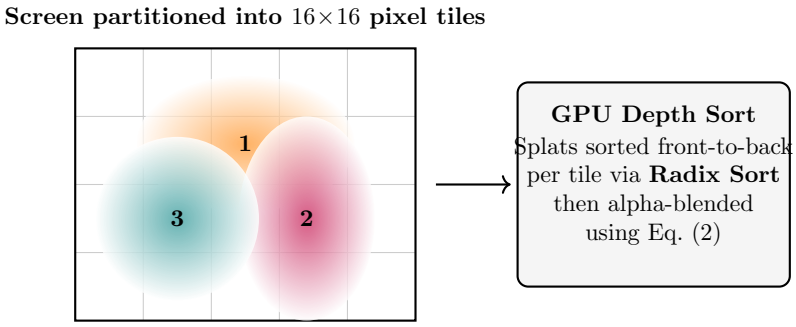

Figure 3: Fast differentiable rasterisation. 3D Gaussians are projected onto the image plane as 2D splats (numbered front-to-back). The screen is divided into 16 × 16-pixel tiles; a GPU Radix sort orders splats by depth within each tile, and alpha blending accumulates pixel colours. The whole pipeline is differentiable, enabling gradient-based optimisation of all Gaussian parameters.

2.5 Anisotropy Metrics

The anisotropy ratio of Gaussian i quantifies how elongated its shape is:

$$\rho_i = \frac{\max(s_{i,0}, s_{i,1}, s_{i,2})}{\min(s_{i,0}, s_{i,1}, s_{i,2})}, \tag{4}$$

where $s_{i,k} = \exp(\hat{s}_{i,k})$ are physical scales obtained by exponentiating the raw optimised log-scale parameters. A ratio near 1 corresponds to a spherical (isotropic) Gaussian; large ratios indicate needle- or disk-like shapes. Two complementary metrics are also defined:

$$\text{Elongation}_i = s_{\text{max},i}/s_{\text{mid},i}, \tag{5}$$

$$\text{Flatness}_i = s_{\text{mid},i}/s_{\text{min},i}, \tag{6}$$

where s_{mid} is the median principal scale. High elongation with low flatness characterises needle-like Gaussians; the reverse characterises disk-like Gaussians; both small characterises isotropic blobs. The ablation study in [3] demonstrates that anisotropic covariance is a critical contributor to visual quality, particularly for fine structures and surface detail.

2.6 The Tanks & Temples Train Scene

The Tanks & Temples benchmark [4] provides large-scale real-world sequences for evaluating 3D reconstruction and novel-view synthesis. The *Train* scene captures a vintage locomotive on a rail track. Its surface properties are geometrically diverse. Rail gravel and ballast produce quasi-random high-frequency texture, so Gaussians there should be small and weakly anisotropic. The broad low-curvature painted metal fuselage should yield large disk-like Gaussians. Rivets, edges, and railings represent elongated 1D structures suited to needle-like Gaussians. Surrounding vegetation is irregular and non-smooth, likely inducing mixed anisotropy. This heterogeneity makes the scene well-suited to studying whether learned anisotropy correlates with underlying scene geometry.

3 Implementation

3.1 Training Setup

We used the publicly available 3DGS implementation of Kerbl et al. [3]. Camera poses and a sparse SfM point cloud were computed with COLMAP [8]. The model was trained for 30,000 iterations with the default loss in equation (3), adaptive densification every 100 iterations, and

opacity resets every 3,000 iterations on a single GPU. The optimised Gaussian cloud was saved as a `.ply` file for downstream analysis. The trained model contains 1,071,147 valid Gaussians (99.9% of 1,072,249 total, with near-zero-scale Gaussians filtered at $s_{\min} > 10^{-6}$). All 301 frames were rendered at 980×545 resolution.

3.2 Anisotropy Analysis Pipeline

Raw log-scale parameters $\hat{\mathbf{s}}$ were exponentiated to obtain physical scales. The three metrics in equations (4)–(6) were computed per Gaussian. Gaussians were categorised by the following thresholds:

- **Needle-like**: Elongation > 5 and Flatness < 2,
- **Disk-like**: Elongation < 2 and Flatness > 5,
- **Blob-like (isotropic)**: Elongation < 2 and Flatness < 2,
- **Mixed**: all remaining Gaussians.

Nearest-neighbour distances and Pearson correlations were computed to assess the spatial organisation of high-anisotropy Gaussians relative to low-anisotropy ones.

3.3 Rendering Quality Evaluation

All 301 rendered frames were compared against ground-truth frames using PSNR and SSIM [9]. A per-frame binary anisotropic mask was generated using local image variance: the top 30% of pixels by local variance (window 15×15) were designated as anisotropic, corresponding to high-frequency textured regions where anisotropic Gaussians are expected to dominate.

4 Results

4.1 Distribution of Anisotropy

Table 1 summarises the distribution of anisotropy ratio, elongation, and flatness across all 1,071,147 valid Gaussians.

Table 1: Summary statistics of Gaussian shape metrics in the trained Train-scene model.

Metric	Count	Mean	Median	Std	P25	P75	P99
Anisotropy Ratio	1,071,147	301.86	27.77	4,158.60	10.08	89.20	3,794.91
Elongation	1,071,147	28.31	4.67	723.16	2.27	11.49	262.82
Flatness	1,071,147	54.02	4.09	982.03	1.92	12.34	551.45

The median anisotropy ratio of 27.77 confirms that the majority of Gaussians are far from isotropic: the typical Gaussian is nearly 28 times more extended in one direction than another. The mean (301.86) is pulled upward by extreme outliers, consistent with the heavy right tail visible in Figure 4. In total, 88.9% of Gaussians have $\rho > 5$ and 75.2% have $\rho > 10$, so strong anisotropy is the norm rather than the exception in a converged 3DGS model.

A log-normal distribution is a natural candidate for ratio data. Fitting to $\log(\rho_i)$ yields $\hat{\mu} = 3.510$, $\hat{\sigma} = 1.656$, implying a log-normal median of $e^{3.510} \approx 33.46$ and mean of $e^{3.510+1.656^2/2} \approx 131.76$. A Kolmogorov–Smirnov test rejects this fit at the 5% level ($p < 0.001$), indicating heavier tails than log-normal. The fit nonetheless serves as a useful reference model, as shown in Figure 5.

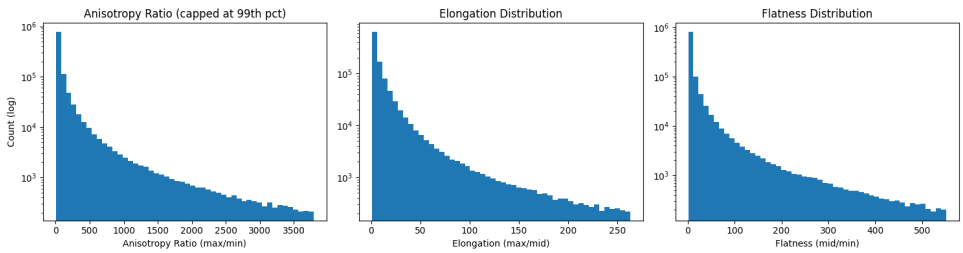

Figure 4: Histograms of anisotropy ratio (capped at 99th percentile), elongation, and flatness distributions, all with log-count y-axis.

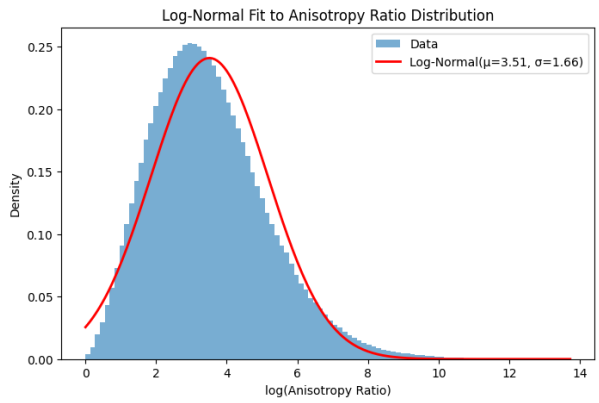

Figure 5: Log-normal fit to the anisotropy ratio distribution, showing $\log(\rho)$ histogram with the fitted normal curve overlaid. The KS test rejects the fit, revealing a heavier-tailed empirical distribution.

4.2 Gaussian Shape Categories

Table 2: Gaussian shape category breakdown for the trained Train-scene model.

Category	Count	Proportion
Needle-like (elongated)	172,455	16.1%
Disk-like (flat)	126,395	11.8%
Blob-like (isotropic)	37,490	3.5%
Mixed	734,807	68.6%

Only 3.5% of Gaussians are truly isotropic blobs. The dominant category is "mixed" at 68.6%, consistent with the scene's many intermediate surfaces. Needle-like Gaussians (16.1%) are expected along the locomotive's edges, wheel spokes, and track rails, while disk-like Gaussians (11.8%) correspond to broad flat surfaces such as the side panels. Figure 6 visualises the 3D spatial distribution of each category. All four categories share the overall scene silhouette, with needle-like and disk-like Gaussians more concentrated in the foreground locomotive region rather than in background vegetation.

4.3 Spatial Organisation of Anisotropy

Vertical stratification. Figure 7 (left panel) shows a bimodal distribution in the Y-coordinate with a sharp spike near $Y = 0$. The top 10% most anisotropic Gaussians are concentrated in a narrow horizontal band at the locomotive's body height rather than distributed throughout the scene.

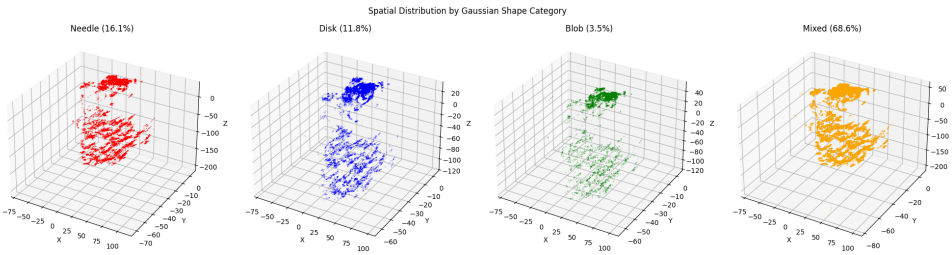

Figure 6: 3D scatter plots of the four Gaussian shape categories (needle in red, disk in blue, blob in green, mixed in orange) in world-space coordinates. Needle- and disk-like Gaussians concentrate in the foreground.

Anisotropy vs. height. The central panel shows that the highest anisotropy values ($\log(1+\rho) > 8$) appear specifically near the rails and wheel base at $Y \approx 0$, the most geometrically complex zone between the locomotive and the track.

Top-down view. The right panel shows the X–Z plane coloured by log-anisotropy. Anisotropy is elevated along the length of the locomotive body and diminishes toward the scene periphery, confirming that the orientation of Gaussian elongation follows the scene geometry.

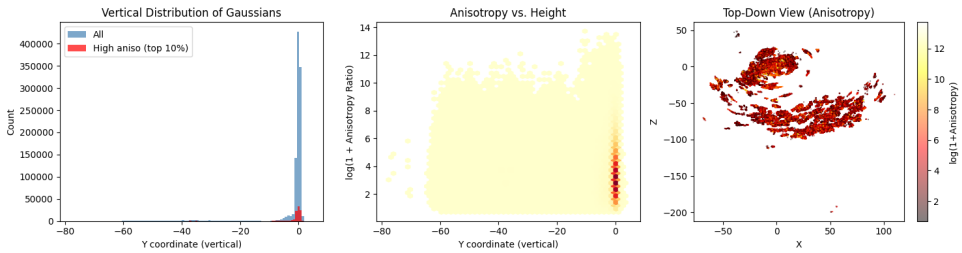

Figure 7: Three-panel spatial analysis: (left) vertical distribution of all Gaussians and the high-anisotropy top 10%; (centre) anisotropy vs. height hexbin; (right) top-down view coloured by log-anisotropy.

Nearest-neighbour clustering test. To test whether high-anisotropy Gaussians are more spatially clustered, mean 5-nearest-neighbour distances were computed on samples of 5,000 Gaussians from each stratum:

$$d_{\mathrm{NN}}(\text{high aniso}) = 0.6161, \quad d_{\mathrm{NN}}(\text{low aniso}) = 0.5771, \quad \text{ratio} = 0.94.$$

High-anisotropy Gaussians are slightly less clustered than low-anisotropy ones. This reflects the fact that elongated Gaussians are spread broadly to cover large surfaces efficiently, whereas tightly packed isotropic Gaussians handle compact detail-rich sub-regions. The Pearson correlation between local point density and anisotropy ratio is $r = -0.018$, confirming negligible linear dependence. However, the correlation between local anisotropy variation (standard deviation of neighbours' ρ values) and ρ is $r = 0.373$, indicating that highly anisotropic Gaussians reside in geometrically heterogeneous neighbourhoods where the optimal primitive shape varies rapidly.

4.4 Rendering Quality

4.4.1 Quantitative metrics

Table 3 summarises rendering quality over all 301 test frames.

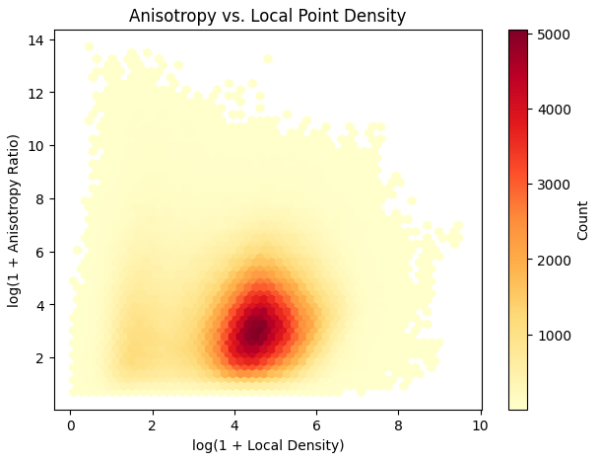

Figure 8: Hexbin scatter of $\log(1+\text{local density})$ vs. $\log(1+\rho)$ for all valid Gaussians, illustrating the near-zero linear correlation between local density and anisotropy ratio.

Table 3: Rendering quality: anisotropic regions (top 30% local variance) vs. isotropic regions, averaged over 301 frames.

Metric	Count	Mean	Std	Min	Max
PSNR (anisotropic regions)	301	22.18 dB	2.15	16.56	28.32
PSNR (isotropic regions)	301	27.61 dB	2.91	17.88	36.05
PSNR gap (iso − aniso)	301	5.42 dB	1.86	0.70	10.91
SSIM (full image)	301	0.864	0.047	0.667	0.959

Anisotropic image regions are on average 5.42 dB worse in PSNR than isotropic regions. The gap is stable across views (std = 1.86 dB) and always positive (minimum = 0.70 dB), confirming a systematic effect rather than a view-specific artefact.

4.4.2 Visual analysis

Figure 9 presents a representative frame (frame 150 of 301). The absolute error map reveals concentrated errors along the locomotive's body, wheel bogies, and rail track, which are precisely the regions identified as anisotropic by the local-variance mask. The error-plus-boundary overlay confirms that error hotspots align tightly with the edges of the anisotropic mask. The bar chart inset shows the 5.42 dB gap (22.18 dB vs. 27.61 dB) directly.

5 Discussion

The results confirm that the converged model encodes a geometrically meaningful anisotropic structure. Needle-like Gaussians (16.1%) concentrate along elongated structures such as rails, railings, and wheel spokes. Disk-like Gaussians (11.8%) cover the broad metallic side panels. High-anisotropy Gaussians are vertically stratified at locomotive height rather than uniformly spread, and local anisotropy variation correlates positively ($r = 0.37$) with anisotropy magnitude, indicating that complex geometry drives extreme elongation.

However, the 5.42 dB PSNR gap shows that capturing anisotropic correlation in the primitive representation does not directly translate into high rendering fidelity in those regions. The current parameterisation, while substantially better than the isotropic alternative (the ablation in [3] shows several dB of degradation when isotropy is enforced), still leaves room for improvement.

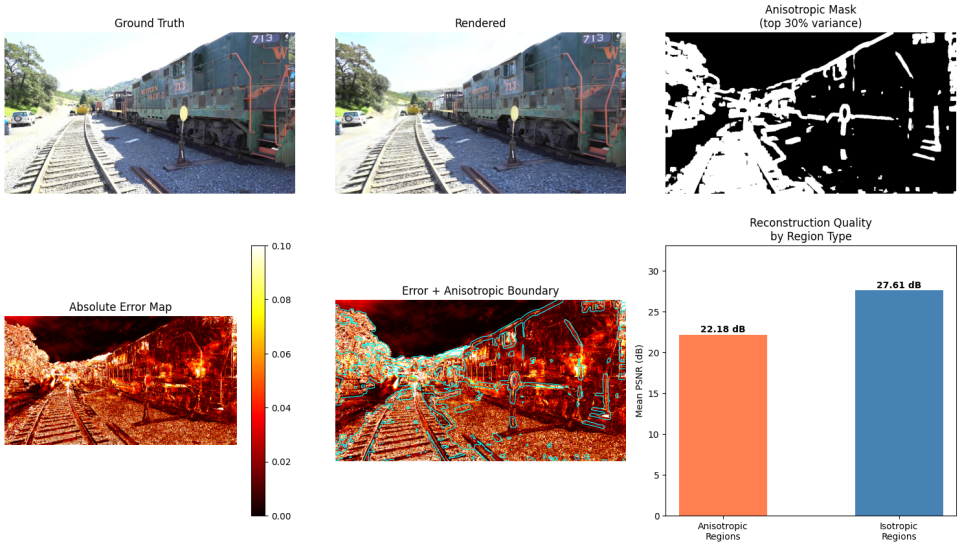

Figure 9: Six-panel figure: (top row) ground truth, rendered image, anisotropic mask; (bottom row) absolute error map, error with anisotropic boundary overlay, bar chart of mean PSNR by region type.

6 Potential Algorithmic Improvements for Anisotropic Regions

The standard adaptive density control (ADC) of Kerbl et al. uses the 2D view-space positional gradient as the sole criterion for cloning or splitting Gaussians (Section 2.3). This gradient is view-dependent and can be small in anisotropic regions observed under only a few camera poses, leaving fine elongated structures under-reconstructed even when the photometric loss remains high.

Scaffold-GS [5] addresses this structurally by organising Gaussians around a sparse voxel grid of anchor points and predicting per-Gaussian attributes, including covariance offsets on-the-fly using a lightweight MLP conditioned on viewing direction and distance. Rather than cloning individual Gaussians, the method grows or prunes anchors based on accumulated neural importance, which is less sensitive to any single viewpoint. Concretely, a new anchor is inserted whenever the maximum accumulated gradient magnitude across all training views of its child Gaussians exceeds a threshold; this multi-view criterion is better suited to detecting under-reconstructed anisotropic regions than the single-frame gradient used by the baseline.

7 Conclusion

This project applied 3D Gaussian Splatting [3] to the Tanks & Temples Train scene to study the relationship between scene anisotropy and model behaviour.

Scene selection (O1). The Train scene provides a diverse testbed through its combination of flat painted metal, elongated edge structures, corroded rivet patterns, and surrounding vegetation, all of which predict qualitatively different optimal Gaussian shapes.

Training and characterisation (O2, O3). A converged 30K-iteration model contains approximately 1.07 million Gaussians. The median anisotropy ratio of 27.77 (with 88.9% of Gaussians having $\rho > 5$ and only 3.5% truly isotropic) demonstrates that anisotropy is the norm. High-anisotropy Gaussians are spatially concentrated at the locomotive's body height and along its length, providing geometric evidence that the model encodes structural information through its primitive shapes.

Algorithmic improvements (O4). Geometry-aware anchor densification proposed by Lu et.

al. [5] to allocate primitives more reliably in anisotropic regions by replacing the single-view gradient criterion of the baseline ADC with a multi-view anchor importance measure

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